

ISURU RAJAPAKSHE

+1(213)643-9972 • rmisur@gmail.com • Los Angeles, CA

 [linkedin.com/in/isuru-p-rajapakshe](https://www.linkedin.com/in/isuru-p-rajapakshe)

 github.com/IsuruPabasara

 isurupabasara.github.io

ABOUT ME

Electrical Engineering graduate with specialization in biomedical research and embedded systems. Experienced in real-time signal processing, device prototyping, and machine learning (PyTorch, MATLAB). Passionate about building intelligent health-tech and automation systems.

SKILLS

- Circuit design(Multisim, OrCAD, Proteus)
- Signal Processing(MATLAB, Python)
- Embedded systems(Zephyr RTOS, Arduino, C)
- Machine Learning(Pytorch, MATLAB)
- Robotics(Kinematics, PID, Trajectory planning)
- Web development(React, NodeJS, CSS/HTML/JS)

EXPERIENCE

Research Assistant | *Intellimed Research Lab, Cal State LA* Dec. 2023 - Current

- Investigating biomarkers for Major Depressive Disorder using spectral analysis (MATLAB / Python) of EEG readings in patients undergoing repetitive Transcranial Magnetic Stimulation (rTMS) therapy.

Instructional Student Assistant | *BOOST Program, Cal State LA* Jun. 2024 - Aug. 2024

- Mentored and led a 5-person team to design a gesture-based music generation device using 3D hand-tracking. Delivered under a 3-month client deadline.

Software engineer | *Insync Information Technologies (Startup), Colombo, Sri Lanka* Jul. 2021 - Jul. 2022

- Lead full-stack development engineer of a team building a network automation tool.
- Utilized the MERN stack (MongoDB-Express-React-Node.js) to develop a functional prototype within 7 months leading to a collaboration with Dialog Axiata PLC.

Electronic engineer intern | *Lanka Electronics, Minuwangoda, Sri Lanka* June. 2019 - Dec. 2019

- Designed robotic arm trajectory paths and implemented welding features in (C#)-based control software.
- Achieved 3mm spatial accuracy in a 6-DOF robotic arm system.

PROJECTS

ChillSpot Finder | Mar. 2024 - May. 2024

- Developed an automated seating system using the nRF52 development kit, integrated with a Kivy app, to detect seat availability based on body heat and guide students to open spots in crowded libraries.

LipRead | Nov. 2023 - Dec. 2023

- Devised a pipeline using Pytorch to predict words spoken in a soundless video using Gated Recurrent Units. Achieved a 66% accuracy on the selected dataset.

Non-Intrusive Real-Time Power Monitor | Feb. 2020 - Jul. 2021

- Contributed to a team project on non-intrusive, real-time household power monitoring by developing the web interface using Firebase and designing power meter circuits in OrCAD.
- Achieved a prediction accuracy of 90% for electric kettle and 95.5% for microwave.

Automated Shade Net for Orchid Plantations | Jan. 2019 - May 2019

- Engineered the core electronics (in Proteus) and custom enclosure (in SolidWorks) for a smart automation system that monitors and controls soil moisture and humidity levels in semi-enclosed orchid plantations.

EDUCATION

Master of Science Electrical Engineering | *CGPA - 4.0 / 4.0* Aug. 2023 - May. 2025

- Electrical Engineering, California State University - Los Angeles

Bachelor of Science of Engineering (Honours) | *CGPA - 3.57 / 4.2* Oct. 2016 - Jun. 2021

- Electronic and Telecommunication Engineering, University of Moratuwa, Sri Lanka

LEADERSHIP AND AWARDS

Biomedical Engineering Society, CSULA | *President* 2025 - Present

2024 Spring CSU Biotech Student Travel Grant Program | *CSIBIOTECH* Jul. 2024

IEEE IAS Student Branch, University of Moratuwa | *Treasurer* 2019 - 2020